SWA-MPPI

Socially-Weighted Alignment with Model Predictive Path Integral Control

Cross-cultural moral alignment of LLMs in logit space

Paper walkthrough (NeurIPS-style)

Technical deep dive

Source: `SWA_MPPI_paper/paper_revised.tex`

Outline

1. Motivation & problem
2. Problem formulation
3. Method pipeline
4. Experiments
5. Mechanisms & robustness
6. Takeaways

Why this problem?

- **Moral preferences are culturally contingent** (Moral Machine: 40M+ judgments across 233 countries).
- **LLMs** inherit predominantly Western / WEIRD training priors (Henrich et al.).
- **MultiTP** (Jin et al.): multilingual trolley benchmark with country-specific human AMCEs—a rigorous testbed for *alignment to local populations*.

Deployment constraint

We want **inference-time** adaptation: **frozen weights**, **no per-culture labelled preference data**, **no reward model training**.

What existing approaches sacrifice

Approach	Limitation for cross-cultural moral alignment
Fine-tuning	Labels + irreversible weight change
Ad-hoc prompting	Weak empirical grounding
Activation steering	White-box, heavy per-model tuning
ARGS / controlled decoding	Trained RM often unavailable
Culture adapters	Scale poorly to 100+ countries

Core insight: *disagreement is signal*

- Within-country moral views are **not monolithic** (age, strata)—supported by survey microdata (WVS Wave 7).
- Build **multiple WVS-grounded personas**; when they **disagree** on a dilemma, the scenario is **culturally contested**.
- That disagreement triggers a **bounded, stochastic** correction in **logit space** (not generation sampling).

Setting: binary moral choice in MultiTP

- Scenario x: forced choice via decision tokens `LEFT` / `RIGHT`.
- Frozen LLM f_θ: logits $\mathrm{z} = [z_\ell, z_r]$, gap $\Delta = z_r - z_\ell$.
- **Human target**: country AMCE vector $\mathrm{h}^{(c)} \in \mathbb{R}^6$ (six moral dimensions).
- **Model estimate**: $\hat{\mathrm{m}}^{(c)}$ from predicted p_{spare} pooled per dimension (MultiTP convention; Utilitarianism via regression on count difference).

Objective

Find logit correction $\delta^*(\mathrm{x}, c)$ so $\hat{\mathrm{m}}^{(c)}$ is closer to $\mathrm{h}^{(c)}$ under **Jensen–Shannon divergence** (primary).

Overview: four stages

1. **Personas** (N=4): 3 WVS age cohorts + 1 utilitarian agent (English system prompts; scenario text stays native-language).
2. **Batched evaluation**: base + N personas in **one** forward pass $\Rightarrow$ per-agent decision gaps δ_i.
3. **Conflict detection**: if inter-agent variance $\nu \geq \tau^{(c)}$, run MPPI; else use consensus only.
4. **PT-weighted MPPI** in logit space + **positional debiasing** (orig vs. lane/group swap).

Persona construction (WVS → natural language)

Eight value dimensions from specific WVS items (e.g. gender egalitarianism Q58P–Q60P, religion Q6P, trust Q43P, . . .).

- Segment respondents into **young / middle / older**; map dimension means to **natural-language descriptors**.
- Fourth persona: **utilitarian** (save more lives)—ablations show it is load-bearing for ranking alignment.

Design principle

Personas reflect **measured population structure**, not author stereotypes (mitigates toxicity risks from naive cultural prompts).

Batched forward pass & gaps

$$\mathsf{z}_{\text{base}}, \mathsf{z}_1, \ldots, \mathsf{z}_N = f_\theta(\mathsf{x}; \varnothing, \mathsf{s}_1, \ldots, \mathsf{s}_N)$$

Per-category logit temperatures T_{cat} scale logits before gaps:

$$\delta_i = \frac{z_{i,r} - z_{i,\ell}}{T_{\text{cat}}}, \quad \bar{\delta} = \frac{1}{N}\sum_i \delta_i, \quad r_i = \delta_i - \delta_{\text{base}}.$$

Reward r_i: how strongly persona i pulls away from the neutral baseline on this scenario.

Adaptive threshold $\tau^{(c)}$ (when to intervene)

- Calibrate on **unlabelled** scenarios using only **inter-agent variance** statistics—**no** human AMCE access.
- Target ≈35% MPPI trigger rate (percentile on calibration variances); main paper uses **20%/80% held-out split** for τ vs. evaluation.
- Robustness: held-out vs. in-distribution calibration yields $|\Delta\mathrm{JSD}| = 0.0006$ (below bootstrap CI ± 0.004).

Prospect Theory utility (per-agent)

Cooperative decomposition with $\lambda_{\text{coop}} = 0.7$:

$$U_i(\epsilon_k) = (1 - \lambda_{\text{coop}})\, v(r_i \tilde{\delta}_k) + \lambda_{\text{coop}}\, v(\bar{r}_{-i}\, \tilde{\delta}_k), \quad \tilde{\delta}_k = \bar{\delta} + \epsilon_k.$$

Classical PT value:

$$v(x) = \begin{cases} x^{\alpha} & x \geq 0 \\ -\kappa |x|^{\beta} & x < 0 \end{cases} \quad (\alpha = \beta = 0.88,\ \kappa = 2.25).$$

Collective objective + MPPI weights

$$U_{\text{total}}(\epsilon_k) = \frac{1}{N}\sum_i U_i(\epsilon_k) - \alpha_{\text{KL}}\frac{\epsilon_k^2}{2\sigma^2} \quad (\alpha_{\text{KL}} = 0.05).$$

$$w_k = \frac{\exp(U_{\text{total}}(\epsilon_k)/\beta)}{\sum_{k'}\exp(\ldots)}, \qquad \delta^* = \sum_k w_k\epsilon_k, \quad \epsilon_k \sim \mathcal{N}(0,\sigma^2).$$

Key design: $\beta = T_{\text{dec}} = 0.5$ aligns MPPI smoothing scale with the final sigmoid decision temperature.

Positional debiasing

- Evaluate each scenario twice: **original** vs. **swap** (LEFT↔RIGHT lanes + Group A↔B in native script).
- Debiased gap: $\delta_i = \left(\delta_i^{\text{orig}} - \delta_i^{\text{swap}}\right)/2$ (mitigates recency / side bias).
- Final: $p_{\text{spare}} = \sigma\left((\bar{\delta} + \delta^*)/T_{\text{dec}}\right)$.

Benchmark, languages, preprocessing

- **MultiTP**: 15 countries, native-language vignettes, 6 moral dimensions (Species, Gender, Age, Fitness, Social Value, Utilitarianism).
- **12 languages** for scenarios; **English** `LEFT/RIGHT` tokens verified as single-token in each model vocabulary.
- Preprocessing: dedup, quality filter (Utilitarianism edge cases), per-category cap, upsampling for stable AMCE—with **sensitivity suite** over configurations.

Models evaluated (six open-weight families)

Model	**Size**	**Mean JSD** Van.→SWA	**Improv.**
Qwen2.5-72B	72B	0.0859 → 0.0536	**+37.6%**
Llama-3.1-70B	70B	0.0937 → 0.0612	+34.7%
Llama-3.1-8B	8B	0.0533 → 0.0436	+22.6%
Qwen2.5-7B	7B	0.0522 → 0.0426	+18.4%
Mistral-Large	126B	0.0457 → 0.0430	+5.8%
Qwen2.5-32B	32B	0.0632 → 0.0633	−0.2%

Mistral: high vanilla alignment ⇒ smaller headroom. Qwen2.5-32B: **logit concentration** ⇒ MPPI weights near uniform (diagnostic, not hidden failure).

Baselines (Llama-3.1-70B)

Method	Mean JSD ↓	Pearson r ↑
SWA-MPPI	**0.0612**	**0.384**
Country-tailored (B1)	0.0680	0.233
PRISM-style (B3)	0.0699	0.148
WVS profile (B2)	0.0737	0.155
Activation steering	0.0877	−0.208
Vanilla	0.0937	−0.026

What creates the gain? (CS + Binary-ARGS)

- **Consensus-shift baselines** on $\bar{\delta}$: best is CS-Clamp; SWA-MPPI still wins by ~ 0.003 JSD $\Rightarrow$ residual explained by **PT asymmetry**, not WVS weighting alone.
- **Binary-ARGS**: ARGS-PT matches SWA-MPPI kernel; ARGS-WVS can *hurt* vs. uniform without PT asymmetry—isolates **Prospect Theory** as the mechanism.

"Anglosphere paradox"

- Largest JSD improvements often in **English-speaking** countries (e.g. AU/GB/US up to $\sim$+72–75% JSD improvement on Qwen2.5-72B).
- Interpretation: models can be **misaligned with local human AMCEs** even when English data is over-represented—RLHF-era priors do not guarantee country-level moral calibration.

Robustness (high level)

- τ: held-out calibration vs. leakage—negligible aggregate JSD gap.
- **Temperatures**: three sweeps, JSD spans < 0.010; defaults conservative vs. best point.
- **Dataset preprocessing**: six configs, aggregate JSD range 0.0034.
- **Qwen2.5-32B**: entropy-aware σ confirms concentrated logits; explains MPPI degradation.

Ablations (Qwen2.5-72B, USA; indicative)

Config	**JSD** $\downarrow$	**Pearson** r $\uparrow$
Full SWA-MPPI	**0.0425**	**0.724**
No MPPI (consensus only)	0.0493	0.650
Without utilitarian persona (N=3)	0.0610	0.441
Linear utility (no PT)	0.0444	0.706

Contributions (summary)

1. First **MPPI in logit space** for cross-cultural moral alignment, training-free.
2. Strong empirical gains on **70B-class** models (+34–38% mean JSD), evaluated across **six** architectures; smaller models +18–23%.
3. Clear **ablation hierarchy** (utilitarian $>$ MPPI $>$ debias $>$ PT).
4. Extensive **robustness** + **mechanistic** evidence (τ, temps, data, CS, ARGS, entropy diagnostic).

Limitations (honest)

- Binary forced-choice / single decision position—extension to multi-option or open-ended is non-trivial.
- WVS personas are a proxy; sparse WVS coverage hurts some countries.
- Hyperparameter validation used synthetic author-annotated proxy labels + transfer checks—real-data validation remains valuable.

Questions?

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